KCC CODE CLUB · GAME MAKERS

Week 3 Lesson Plan

Loops and Randomness

Big idea: If you're about to do the same thing twenty times, don't. Tell the computer to repeat it. And if you want the game to surprise you, ask it for a random number.

At a glance

New concepts	Iteration (`repeat`, `forever`), randomness, nested loops
New blocks	`repeat n`, `forever`, `repeat until`, `wait n secs`, `pick random a to b`
Class activity	Rain
Project checkpoint	One object falls from a random position at the top, over and over
Homework	A second falling object; random falling speed

Before the class

- Print the Week 3 handout.
- Skeleton `Sky Catcher — end of Week 2` ready for anyone who missed a week.

0:00–0:05 — Show & Tell

Homework demos, two of them. By now you should get three or four hands, so choose. If someone has done a bonus challenge, get them up — bonus work being celebrated is what makes the *next* student attempt one.

0:05–0:15 — Live demo: repeat, forever, random

Frame it before you touch the keyboard. Clap a pattern — *clap, clap, stomp* — four times, wearily, then write it on the board as `repeat 4 [clap, clap, stomp]` and change the 4 to a 10. Twelve actions became thirty by editing one character. Thirty seconds, and the nested loop later lands much more easily for it.

Build live:

1. `when green flag clicked` → `repeat 10 [ move 10 steps, wait 0.1 seconds ]`. Run it. It crawls across the stage.
2. Take out the `wait`. Run again. It teleports — too fast to see. Put it back. The point: **the wait is what makes it visible**, and this is why their falling object will "not work" later.
3. Swap `repeat 10` for `forever`. Note there's no bump at the bottom of a `forever` block — nothing can come after it, because it never ends. Ask them why.
4. Introduce `pick random 1 to 10`. Drop it inside `move [pick random 1 to 10] steps`. Run it several times. Different every time.
5. Show `go to x: [pick random -240 to 240] y: 180`. Run it repeatedly. It's jumping around the top of the stage — which is exactly the falling-object spawn they're about to build.

6. **Deliberate mistake:** put the `wait` *outside* the loop instead of inside. Run it. Be confused for a moment, then narrate finding it.

0:15–0:30 — Class activity: Rain

This is the dress rehearsal for the project build, deliberately.

```
when green flag clicked                                   EVENTS
forever                                                   CONTROL
    go to x: (pick random -240 to 240) y: 180             MOTION
    repeat 20                                             CONTROL
        change y by -18                                   MOTION
        wait 0.02 seconds                                 CONTROL
```

A raindrop that appears at a random point along the top and falls to the bottom, forever. Get them to change the numbers: fewer repeats and a bigger step is a faster drop.

Fast finishers: ask them to make it look like rain by duplicating the sprite three or four times. Duplicating a sprite copies its scripts, which is a nice thing for them to discover. After that, the Dancing Sprite bonus challenge.

0:30–0:55 — Project build: things fall from the sky

Goal: **"By the end of today, something falls from the sky in your game, over and over."**

This is the week the project starts looking like a game, and they feel it.

1. Add a new sprite that matches the theme — the mango, the ball, the star.
2. Shrink it if needed (`change size by` or the size field in the sprite pane).
3. Build the falling script. Same shape as Rain:

```
when green flag clicked                                   EVENTS
forever                                                   CONTROL
    go to x: (pick random -220 to 220) y: 170             MOTION
    repeat 20                                             CONTROL
        change y by -18                                   MOTION
        wait 0.02 seconds                                 CONTROL
```

4. **Tune it.** Make everyone change the numbers until the speed feels playable with their player. Too fast and week four's catching will be impossible; too slow and it's boring. This is real game design and they should spend five minutes on it.

Note that the falling object doesn't yet notice the player at all — you can "catch" it and nothing happens. Say that out loud and promise it for next week. The anticipation is useful.

Save.

0:55–1:00 — Wrap

Save, back up, homework brief.

Homework

1. **A second thing to catch.** Add another falling sprite with a different look. (Hint: right- click the first one and duplicate it, then change its costume — the script comes with it.)
2. **Unpredictable speed.** Use `pick random` so each drop falls at a different speed. (Hint: the `wait` inside the falling loop can take a random number.)

Bonus challenges

1. Draw a shape with `repeat` and `turn`. A square is 4 and 90. What makes a triangle? A hexagon? What happens with `repeat 36 [ move 10, turn 10 ]`?
2. Make the falling object spin as it falls.
3. **Dancing Sprite.** Pick a sprite with a walk cycle. `forever` → `next costume`, `wait 0.2 seconds`, `move 5 steps`. Tune the wait until the walk looks right, and add `if on edge, bounce` so it doesn't escape.
4. Use `repeat until` instead of `forever` so the object stops after ten drops. (You'll need a way to count, which is properly a week-five problem. See how far you get.)

Common bugs this week

Symptom	Cause
Object flickers at the top and nothing else	No `wait` inside the falling loop, so a whole drop happens in one frame
Object falls once then stops	The `go to` and the `repeat` aren't wrapped in a `forever`
Object falls at an angle	`change x by` in the loop as well as `change y by`
It falls upward	`change y by 18` instead of `-18`. Positive y is up
Object goes below the stage and reappears oddly	Too many repeats for the step size; the sprite is clamped at the edge
Scratch freezes / the browser tab hangs	A `forever` with no `wait` inside it at all. Add a `wait 0.01`, always

That last one is worth warning about pre-emptively. A `forever` with no wait will lock the tab and they'll lose unsaved work. Say "every forever needs a wait somewhere inside it" and repeat it all term.

Notes for the second volunteer

The nested loop in the falling script (a `repeat` inside a `forever`) is the first genuinely abstract thing in the course. Some children will build it correctly and still not be able to explain it. That's fine at this stage — ask them to predict what happens if you change a number, and let understanding follow the doing.

Links

- Find the bug — Raspberry Pi Foundation
- Skeleton: `Sky Catcher — end of Week 3` in the club studio